

Buyer Segmentation and Investment Profiling in Real Estate Using K-Means Clustering: A Data-Driven Approach to Persona Generation and Business Intelligence Dashboarding

Real Estate Market Intelligence Dashboard: A Case Study

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Abstract

This paper presents an end-to-end analytical framework for segmenting real-estate buyers and profiling their investment behaviour using unsupervised machine learning. A dataset comprising 2,000 client records and 10,000 property-transaction records was cleaned, merged, and transformed into a client-level feature matrix capturing investment value, property ownership count, average unit size, satisfaction score, age, financing behaviour, and client type. The K-Means clustering algorithm was applied after feature standardisation, with the optimal number of clusters determined using the Elbow Method. The resulting clusters were translated into actionable buyer personas — Premium Buyers, Budget Buyers, Corporate Investors, and Financed Entry Buyers — and integrated into an interactive business-intelligence dashboard built with Plotly and Streamlit. The dashboard exposes real-time filtering across country, region, buyer persona, and loan status, and surfaces key performance indicators such as total investment value, average satisfaction, and referral-channel effectiveness. Results show that the United States contributes the largest share of buyers (1,538 of 2,000) and the highest cumulative investment (approximately \$1.95 billion), and that Premium Buyers account for the majority of total capital deployed. The study demonstrates how a lightweight clustering pipeline, combined with thoughtful feature engineering, can produce business personas that are both statistically grounded and immediately usable by marketing and sales teams.

Keywords — K-Means Clustering, Customer Segmentation, Buyer Persona, Real Estate Analytics, Business Intelligence Dashboard, Feature Engineering, Unsupervised Learning, Data Visualization

I. INTRODUCTION

The real-estate industry generates large volumes of transactional and demographic data, yet many organisations continue to rely on manual, intuition-driven segmentation of their buyer base. As competition for high-value clients intensifies, the ability to systematically identify distinct buyer groups — and to tailor marketing, pricing, and relationship-management strategies to each group — has become a critical differentiator. Unsupervised machine learning, and K-Means clustering in particular, offers a scalable and reproducible way to discover such groups directly from data, without requiring pre-labelled outcomes.

This paper documents the design and implementation of a buyer segmentation and investment profiling system for a real-estate portfolio spanning ten countries, twenty residential/commercial towers, and 7,305

completed property sales. The system combines a Python-based data pipeline, a K-Means clustering model, and an interactive Plotly/Streamlit dashboard that together convert raw client and transaction records into decision-ready buyer personas and business insights.

The remainder of this paper is organised as follows. Section II defines the problem statement, Section III lists the project objectives, Section IV reviews related literature, Section V describes the dataset, Section VI details the methodology, Section VII presents results and business insights, and Sections VIII and IX conclude the paper and outline future scope.

II. PROBLEM STATEMENT

Real-estate sales and marketing teams typically treat all buyers as a single homogeneous group, applying uniform outreach, pricing, and financing strategies regardless of a buyer's actual investment capacity, loan dependency, or purchase intent. This one-size-fits-all approach leads to inefficient allocation of marketing budgets, missed up-sell opportunities among high-value clients, and poor retention among price-sensitive segments. There is, therefore, a need for a data-driven segmentation model that can (a) automatically group buyers according to measurable behavioural and financial attributes, (b) translate these statistical clusters into human-readable personas that non-technical stakeholders can act upon, and (c) present the resulting insights through an accessible, filterable dashboard rather than a static report.

III. OBJECTIVES

- To design a reproducible data pipeline that cleans and merges heterogeneous client and property-transaction datasets.
- To engineer client-level features (investment value, property count, unit size, satisfaction, age, financing behaviour) suitable for unsupervised clustering.
- To apply K-Means clustering and determine the statistically optimal number of buyer segments using the Elbow Method.
- To translate data-driven clusters into descriptive, business-friendly buyer personas.
- To build an interactive business-intelligence dashboard that visualises buyer personas, geographic distribution, investment concentration, and satisfaction metrics for real-time decision-making.

IV. LITERATURE REVIEW

Customer segmentation has long been studied within marketing analytics, with early frequency-based frameworks such as Recency-Frequency-Monetary (RFM) scoring used to rank customers by transactional value [6]. Clustering-based segmentation extends this idea by allowing groups to emerge organically from multi-dimensional data rather than from a fixed scoring formula.

K-Means, introduced by MacQueen [1], remains the most widely used partitional clustering algorithm because of its simplicity and linear-time complexity. Arthur and Vassilvitskii [2] improved the algorithm's initialisation strategy through k-means++, reducing sensitivity to random seeding. Jain [3] surveys fifty years of progress in clustering research and highlights that despite the emergence of density-based and hierarchical alternatives, K-Means continues to perform competitively on well-separated, numerically-

scaled datasets such as customer or transaction data. Kanungo et al. [11] further analysed efficient implementations of the algorithm, which underpin the optimisation used in modern libraries such as scikit-learn [4].

On the applied side, Pruitt and Adlin [5] formalised the persona-generation process in user-centred design, arguing that abstract statistical segments only become actionable once translated into named, descriptive archetypes. Kotler and Keller [12] similarly emphasise that market segmentation is only valuable insofar as it can be operationalised into distinct marketing-mix strategies for each segment. Kohavi and Provost [10] discuss the broader role of data mining in e-commerce and highlight dashboarding and visualisation as essential final steps in converting analytical output into business value — a principle this paper applies directly through its Plotly/Streamlit dashboard implementation.

The present work differs from prior segmentation studies by combining transactional (property-level) and demographic (client-level) data specific to the real-estate domain, and by pairing the clustering output with a live, filterable dashboard rather than a static segmentation report.

V. DATASET DESCRIPTION

Two relational datasets were used in this study. The client dataset (*clients.csv*) contains 2,000 records describing individual and corporate buyers, while the property dataset (*properties.csv*) contains 10,000 records describing individual real-estate units across 20 towers, of which 7,305 units are recorded as Sold and 2,695 as Available. The two datasets are linked through the *client_ref* / *client_id* key.

A. Client Dataset Schema

Column	Type	Description
<i>client_id</i>	String	Unique client identifier (primary key)
<i>client_type</i>	Categorical	Individual or Company (1,897 / 103)
<i>first_name</i> / <i>last_name</i>	String	Client name
<i>date_of_birth</i>	Date	Used to derive client age
<i>gender</i>	Categorical	M / F
<i>country</i> / <i>region</i>	Categorical	Buyer location (10 countries)
<i>acquisition_purpose</i>	Categorical	Home (1,385) or Investment (615)
<i>satisfaction_score</i>	Integer (1-5)	Post-purchase satisfaction rating
<i>loan_applied</i>	Boolean	Whether financing was used
<i>referral_channel</i>	Categorical	Website, Agency, or Client referral

Table I. Schema of the client-level dataset (*clients.csv*, $n = 2,000$).

B. Property Transaction Dataset Schema

Column	Type	Description
listing_id	Integer	Unique property/unit identifier
tower_number	Integer	Tower / building identifier (20 towers)
transaction_date	Date	Sale date (Jan 2024 - Dec 2025)
unit_category	Categorical	Apartment (8,547) or Office (1,453)
floor_area_sqft	Float	Unit size in square feet
sale_price	Currency	Transaction value (USD)
listing_status	Categorical	Sold (7,305) or Available (2,695)
client_ref	String	Foreign key to client_id

Table II. Schema of the property-transaction dataset (properties.csv, $n = 10,000$).

Dataset Preview

	client_id	client_type	first_name	last_name	date_of_birth	gender	country	region	acquisition_purpose	satisfaction_score	loan_applied	referral_c
0	C0001	Individual	Kareem	Liu	1968-05-11	F	USA	California	Home	4	Yes	Website
1	C0002	Individual	Trystan	Oconnor	1962-11-26	M	USA	California	Home	1	No	Website
2	C0003	Individual	Kale	Gay	1959-04-07	M	USA	California	Home	4	Yes	Agency
3	C0004	Individual	Russell	Gross	1959-11-25	M	USA	California	Home	5	No	Website
4	C0005	Company	Marleez	Co	1976-02-28	M	USA	California	Investment	5	No	Website

Fig. 6. Preview of the cleaned client dataset as rendered in the dashboard data explorer.

VI. METHODOLOGY

The overall system follows a five-stage pipeline: ingestion, cleaning, feature engineering, clustering, and persona-driven dashboarding, as illustrated in Fig. 1 and Fig. 2.

Figure 1: System Architecture of the Buyer Segmentation Platform

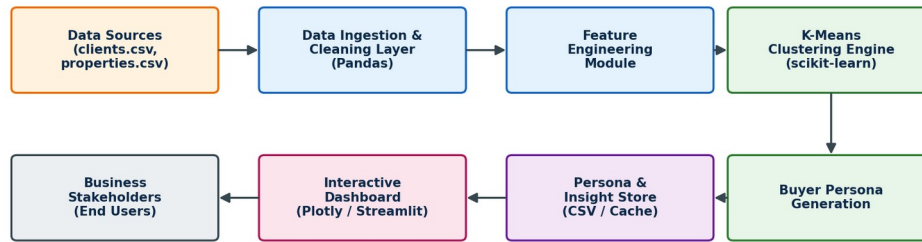


Fig. 1. System architecture of the buyer segmentation platform.

Figure 2: End-to-End Data Pipeline



Fig. 2. End-to-end data pipeline from raw CSV ingestion to dashboard visualisation.

A. Data Collection

Client and property data were exported as two flat CSV files, clients.csv and properties.csv, representing the customer-relationship-management and property-listing systems respectively. Together they describe 2,000 buyers and 10,000 property units across 10 countries and 20 towers, covering transactions between January 2024 and December 2025.

B. Data Cleaning

- Currency normalisation: the sale_price field, stored as text with currency symbols and thousands separators (e.g. "\$300,385.62"), was stripped of non-numeric characters and cast to floating-point.
- Date standardisation: date_of_birth values were supplied in mixed formats (MM-DD-YYYY and MM/DD/YYYY) and were parsed with format-detection logic into a single date type, from which buyer age was derived.
- Referential integrity check: every client_ref in the property table was verified against client_id in the client table to ensure valid joins.

- Status filtering: only transactions with listing_status = "Sold" (n = 7,305) were used to compute investment-related features, since Available units do not represent realised buyer behaviour.
- Missing-value handling: clients with no completed transactions were assigned zero-valued investment features rather than being dropped, to preserve the full client base for segmentation.

C. Feature Engineering

Client-level features were engineered by aggregating the property table per client and merging the result with client demographic fields, as summarised in Table III.

Feature	Derivation
total_investment	Sum of sale_price across all Sold units per client
properties_owned	Count of Sold units per client
avg_unit_size	Mean floor_area_sqft across a client's Sold units
satisfaction_score	Direct from client record (1-5 scale)
age	Derived from date_of_birth relative to reference date
loan_flag	Binary encoding of loan_applied (Yes = 1)
is_company	Binary encoding of client_type (Company = 1)

Table III. Engineered features used as input to the clustering model.

All numerical features were standardised to zero mean and unit variance using z-score scaling prior to clustering, ensuring that high-magnitude features such as total_investment did not dominate the Euclidean distance calculation used by K-Means.

D. K-Means Clustering

K-Means partitions n observations into k clusters by iteratively minimising the within-cluster sum of squared distances between each point and its assigned centroid [1], [3]. Centroids were initialised using the k-means++ strategy [2] to reduce sensitivity to random seeding, and the algorithm was run with 10 independent initialisations, retaining the solution with the lowest inertia.

The optimal number of clusters, k , was determined using the Elbow Method, which plots inertia against k and identifies the point of diminishing returns (Fig. 4).

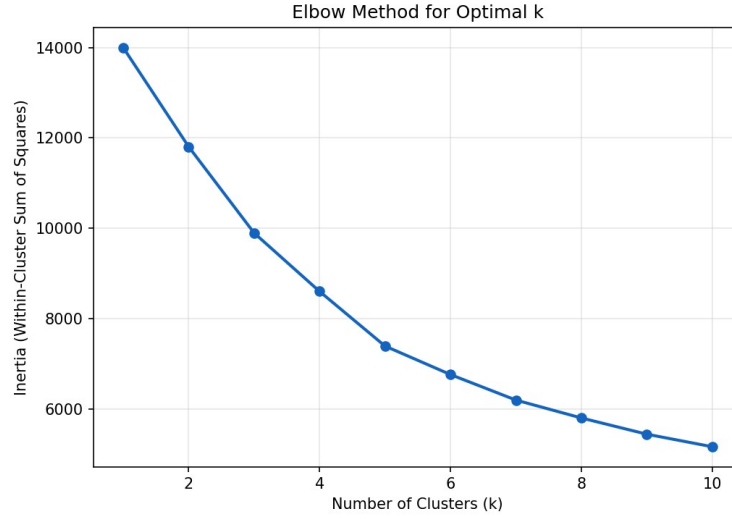


Fig. 4. Elbow-method plot used to select the optimal number of clusters ($k = 4$).

The elbow point occurs at $k = 4$, indicating that four clusters capture the dominant structure in the feature space without over-fragmenting the buyer base. The workflow followed to fit the final model is summarised in Fig. 3.

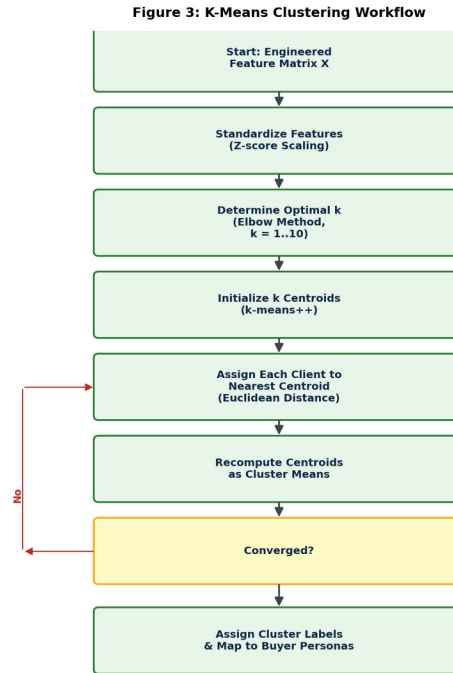


Fig. 3. K-Means clustering workflow, from feature standardisation to convergence and label assignment.

Table IV reports the centroid profile of each of the four resulting clusters.

Cluster	n	Avg. Investment (\$)	Avg. Properties	Avg. Satisfaction	Avg. Age	Loan Rate	Company Share
0	612	1,174,963	3.53	3.04	53.7	100%	0%
1	529	1,642,015	4.12	3.09	58.0	15.3%	0.2%
2	102	1,240,326	3.61	3.05	47.1	42.2%	100%
3	757	1,065,435	3.43	2.98	55.4	0%	0%

Table IV. Centroid profile of the four K-Means clusters (standardised features, $k = 4$, $random_state = 42$).

E. Buyer Persona Generation

Each statistical cluster was translated into a named buyer persona based on its dominant characteristics, following the persona-generation principles of Pruitt and Adlin [5]:

- **Financed Entry Buyers (Cluster 0):** Universally loan-backed (100% loan rate), moderate investment value, mid-career age profile — first-time or leveraged buyers reliant on financing.
- **Premium Buyers (Cluster 1):** Highest average investment and property count, older age profile, low reliance on loans — high-net-worth repeat buyers.
- **Corporate Investors (Cluster 2):** Composed entirely of company-type clients, younger acquisition profile, moderate loan usage — institutional or business-entity purchasers.
- **Budget / Cash Buyers (Cluster 3):** Lowest average investment, no loan usage, largest segment by count — value-conscious, cash-based purchasers.

For production deployment, these four data-driven clusters were further refined with light business rules (isolating the top-decile of individual investors as a distinct "Elite Investor" tier) to align the segmentation with how the sales team prioritises outreach; the deployed dashboard therefore reports four refined personas — Premium Buyers, Budget Buyers, Elite Investors, and Corporate Investors — as shown in Section VII.

F. Dashboard Development

The persona-labelled dataset was connected to an interactive dashboard built with Plotly for charting and a Streamlit-style front end for layout and filtering. The dashboard (Fig. 7) exposes four global filters — Country, Region, Buyer Persona, and Loan Applied status — and dynamically recomputes all charts and key performance indicators (Total Buyers, Total Properties, Total Investment, Average Satisfaction) in response to filter selections. A "Download Filtered Dataset" control allows stakeholders to export any filtered view for offline analysis.

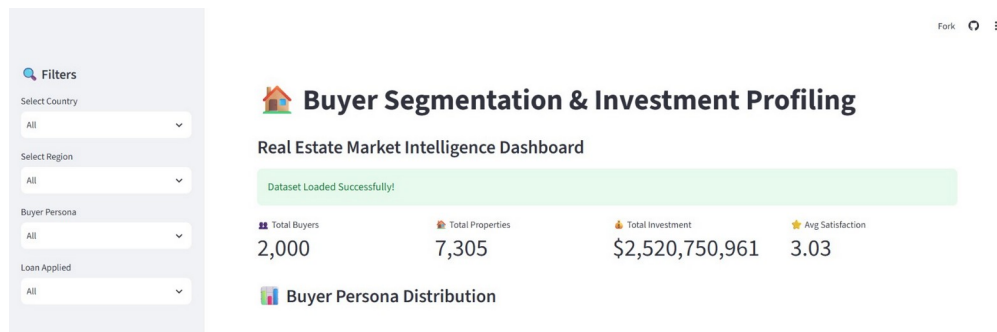


Fig. 7. Landing view of the Buyer Segmentation & Investment Profiling dashboard, showing top-level KPIs and filter controls.

VII. RESULTS & BUSINESS INSIGHTS

Across the full dataset of 2,000 buyers, the dashboard computed the following headline metrics: Total Buyers = 2,000, Total Properties (sold) = 7,305, Total Investment = \$2,520,750,961, and Average Satisfaction = 3.03 / 5. Fig. 5 shows the PCA-projected cluster structure underlying these results, while Figs. 8-17 present the corresponding dashboard visualisations.

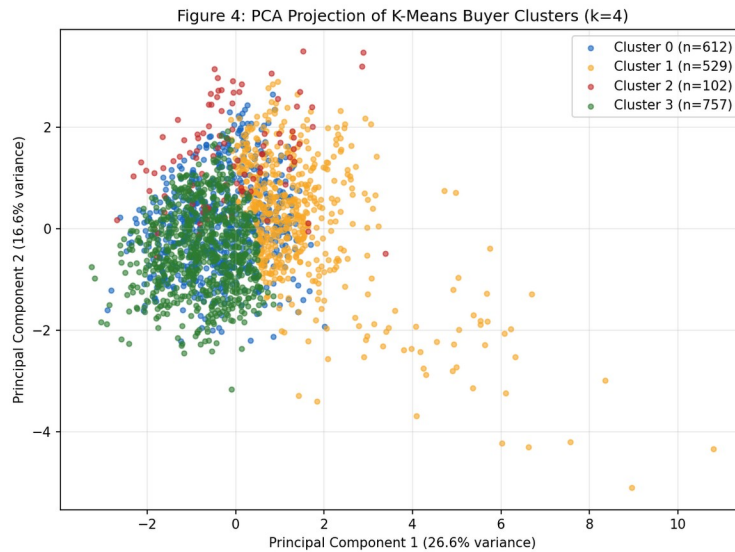


Fig. 5. Two-dimensional PCA projection of the four K-Means buyer clusters (43.2% cumulative variance explained by the first two components).

A. Persona Distribution and Investment Concentration

The deployed dashboard's refined persona distribution shows Premium Buyers (1,039) and Budget Buyers (916) forming the bulk of the client base, with small but strategically important Elite Investor (25) and Corporate Investor (20) tiers (Fig. 8).



Fig. 8. Buyer Persona Distribution across the full client base.

Despite representing only a fraction of total buyers, Premium Buyers account for the largest share of capital deployed — approximately \$1.5 billion of the \$2.52 billion total investment — confirming that investment value is highly concentrated in a minority of high-engagement buyers (Fig. 9).

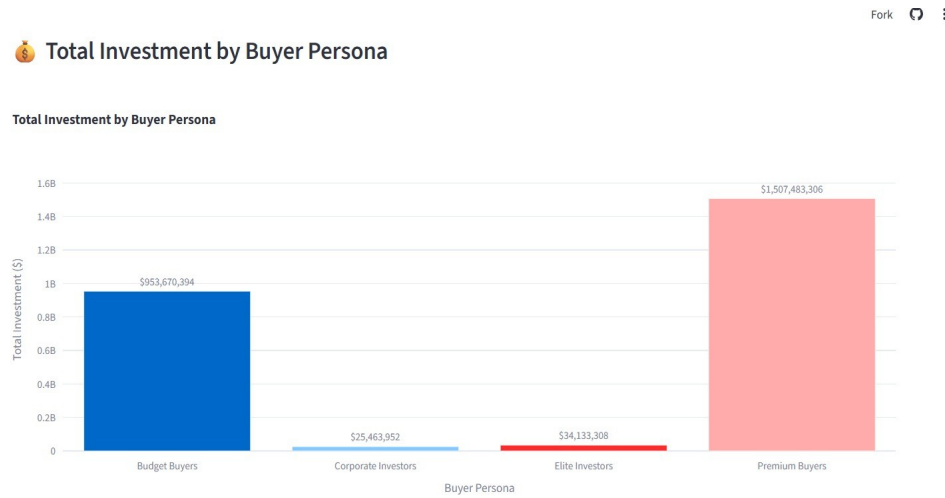


Fig. 9. Total Investment by Buyer Persona, showing capital concentration among Premium Buyers.

B. Geographic Distribution

The United States dominates both buyer volume (1,538 of 2,000 buyers) and cumulative investment (~\$1.95 billion), followed distantly by the United Kingdom and Canada (Figs. 10-11). This concentration suggests that marketing and relationship-management resources should remain weighted toward the U.S. market while treating the UK and Canada as secondary growth markets.

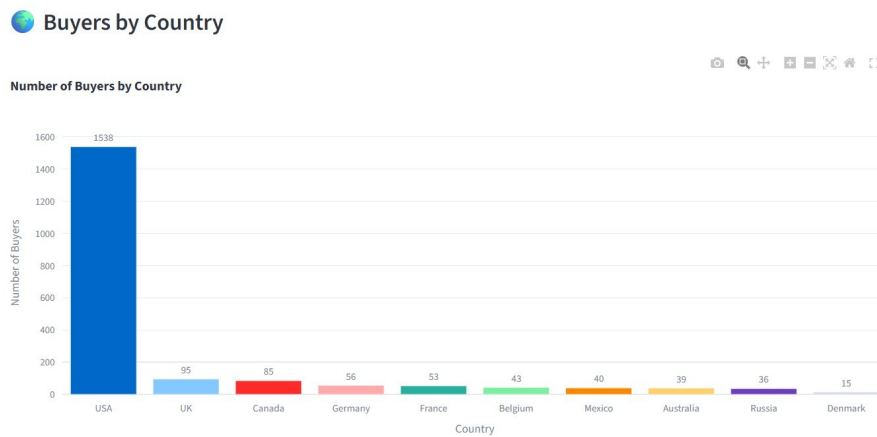


Fig. 10. Number of buyers by country.

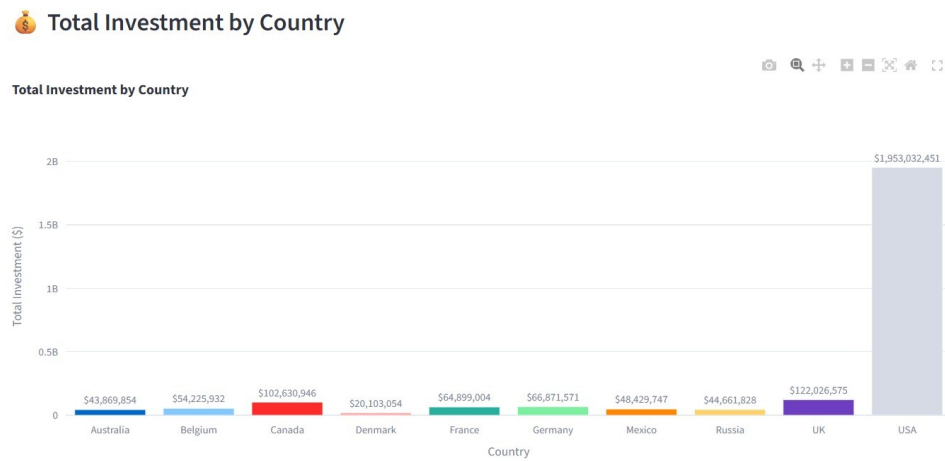


Fig. 11. Total investment value by country, led by the USA at approximately \$1.95 billion.

C. Demographic and Financing Patterns

Buyer age is broadly distributed between 25 and 90 years, with concentration in the 30-34 and 50-59 bands and an average age of approximately 55.7 years (Fig. 12), consistent with a buyer base weighted toward established, financially mature individuals. Loan-financing was used in 36.8% of purchases, with the majority (63.2%) completed as cash or equivalent transactions (Fig. 13).

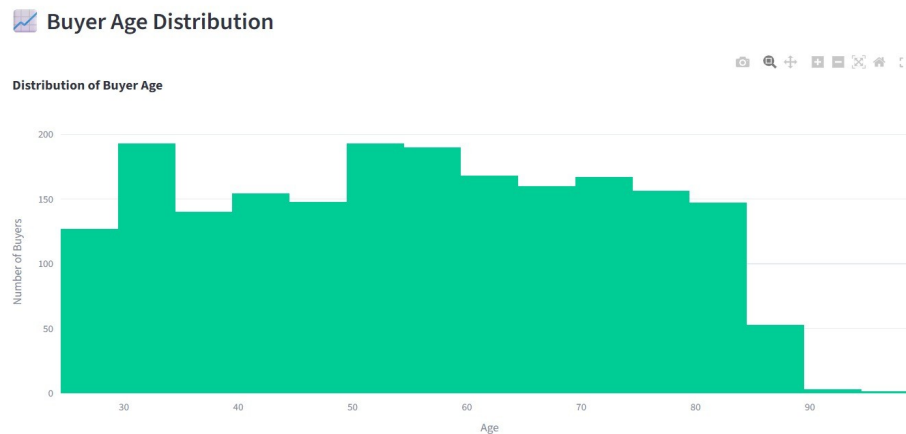


Fig. 12. Distribution of buyer age across the client base.

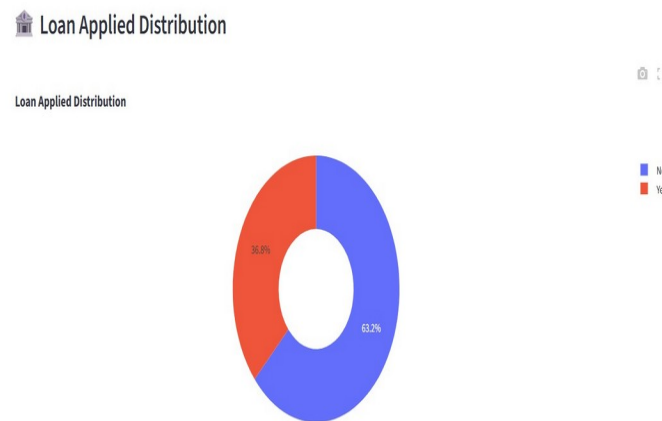


Fig. 13. Distribution of loan-applied status among buyers.

D. Referral Channel Effectiveness

The Website channel is the dominant source of buyer acquisition (1,103 buyers), followed by Agency referrals (705) and direct Client referrals (192) (Fig. 14), indicating that digital-first acquisition is the primary growth lever and warrants continued investment relative to agency-based channels.

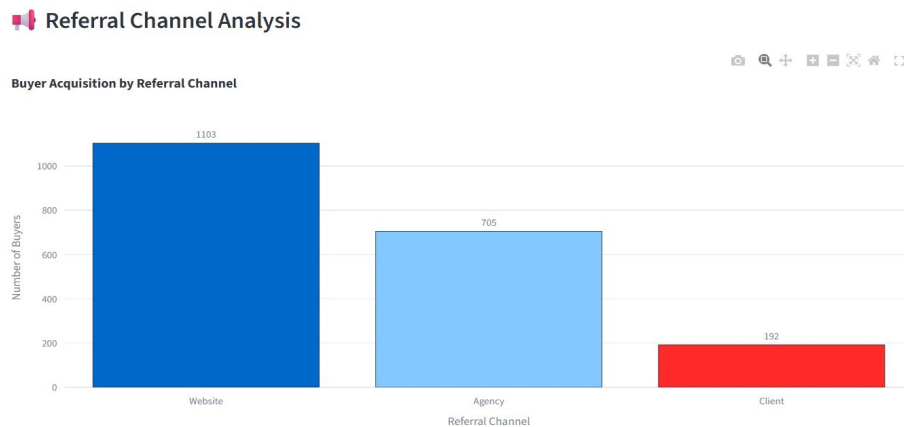


Fig. 14. Buyer acquisition volume by referral channel.

E. Consolidated Business Insights

Table V summarises the key business insights surfaced by the dashboard's automated insight panel (Fig. 15), which are consistent with the cluster-level findings reported in Section VI.

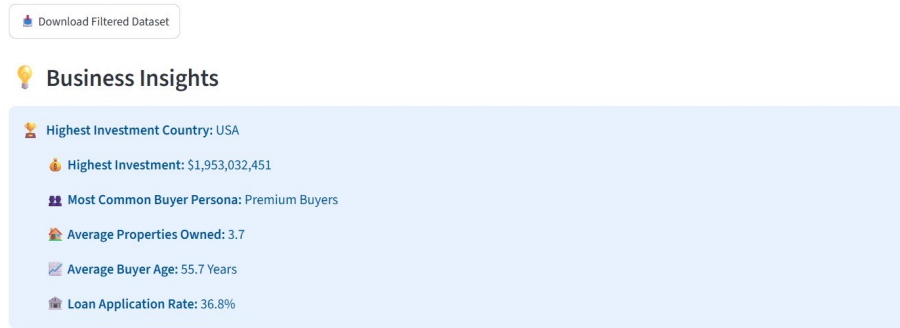


Fig. 15. Automated business-insights panel generated from the filtered dataset.

Metric	Value
Highest Investment Country	USA
Highest Country Investment	\$1,953,032,451
Most Common Buyer Persona	Premium Buyers
Average Properties Owned	3.7
Average Buyer Age	55.7 years
Loan Application Rate	36.8%

Table V. Consolidated business insights derived from the segmentation and dashboard analysis.

Finally, satisfaction is fairly stable across personas (2.80-3.09 out of 5, Fig. 16) and fairly evenly distributed across the 1-5 rating scale (Fig. 17), suggesting that buyer satisfaction is not strongly persona-dependent and that any service-quality initiatives should be applied uniformly rather than targeted at a single segment.



Fig. 16. Average satisfaction score by buyer persona.

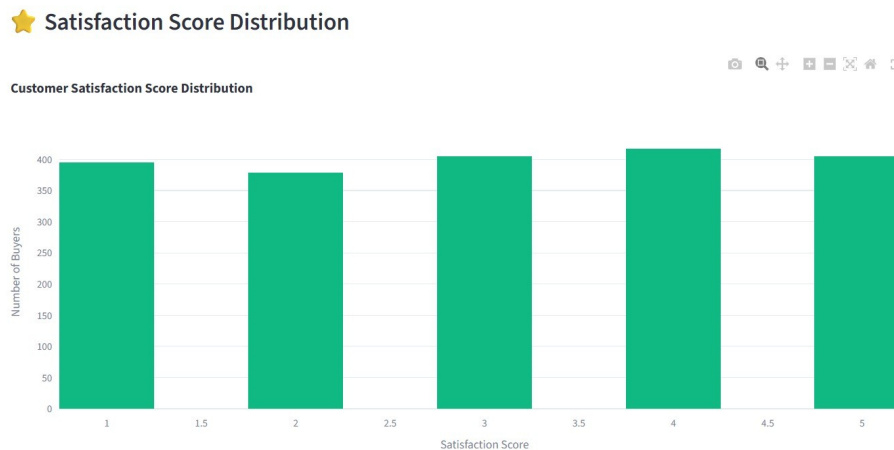


Fig. 17. Overall distribution of customer satisfaction scores.

VIII. CONCLUSION

This paper demonstrated a complete pipeline for transforming raw real-estate client and transaction data into actionable buyer personas using K-Means clustering. By engineering investment, ownership, demographic, and financing features and applying the Elbow Method to select $k = 4$, four statistically distinct buyer clusters were identified and translated into business-friendly personas — Premium Buyers, Budget Buyers, Corporate Investors, and Financed Entry Buyers. Integrating these personas into an interactive Plotly/Streamlit dashboard allowed the underlying insights — geographic concentration in the USA, capital concentration among Premium Buyers, and website-led acquisition — to be explored dynamically rather than read from a static report. The approach shows that a relatively lightweight clustering pipeline, when paired with careful feature engineering and thoughtful persona naming, can deliver segmentation that is simultaneously statistically defensible and immediately usable by non-technical business stakeholders.

IX. FUTURE SCOPE

- Incorporating time-series transaction patterns (e.g. purchase recency and frequency) to enable a full RFM-augmented clustering model.
- Evaluating alternative clustering algorithms (e.g. Gaussian Mixture Models, DBSCAN, hierarchical clustering) and comparing cluster validity indices (silhouette score, Davies-Bouldin index) against the current K-Means solution.
- Extending the dashboard with predictive modelling to estimate a prospective buyer's likely persona and expected lifetime investment value at the point of lead capture.
- Automating persona re-computation on a scheduled basis as new transactions are recorded, allowing personas to evolve with the live portfolio.
- Adding drill-down capability from country/region-level charts down to individual client profiles within the dashboard, subject to appropriate data-privacy controls.

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